Reliability Evaluation Report of the ST33K1M5A

RER222079914

New product

General Information	
Commercial Product	ST33K1M5A
Product Line	K4A082
Product Description	32-bit embedded secure element and embedded SIM
Package	TSSOP20 6.5x4.4 mm VFDFPN8 wettable flank, 5 × 6 mm UFQFPN32 wettable flank, 5 x5 mm
Silicon Technology	CMOSE40

Traceability	
Diffusion Plant	Crolles 300 12
Assembly Plant	ST SHENZHEN - CHINA, SC UTL1 - UTAC Thai Limited

Reliability Assessment	
Pass	X
Fail	

Release	Date	Author	Function
1.7	Feb 13th, 2025	Fabien FRIESS	CS Product Qualification and Reliability Engineer

DOCUMENT ACTORS:

Name	Function	Location	Date
Nicolas DUMONT	CS Product Qualification and Reliability Manager	Rousset	Feb 13th, 2025
Mickael DENAIS	CS Division Quality and Reliability Manager	Rousset	Feb 13th, 2025

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RELIABILITY EVALUATION OVERVIEW

- **OBJECTIVE**

The aim of this report is to present the results of the die-oriented and package-oriented reliability evaluation of the ST33K1M5A secure microcontroller produced in Crolles 300, France and assembled in TSSOP20 at ‘ST SHENZHEN -CHINA’ assembly plant, VFDFPN8 5x6 WF and UFQFPN32 5x5 WF packages at ‘SC UTL1 - UTAC Thai Limited’ assembly plant.

- **CONCLUSION**

All reliability tests have been completed with positive results.

- Die reliability trials performed on the ST33K1M5A manufactured in eSTM40 at Crolles 300.
- Package reliability trials performed on the ST33K1M5A assembled in TSSOP20 package at ‘ST SHENZHEN -CHINA’ assembly plant.
- Package reliability trials performed on the ST33K1M5A assembled in VFDFPN8 5x6 WF package at ‘SC UTL1 - UTAC Thai Limited’ assembly plant.
- Package reliability trials performed on the ST33K1M5A assembled in UFQFPN32 5x5 WF package at ‘SC UTL1 - UTAC Thai Limited’ assembly plant.

As a consequence, the ST33K1M5A product diffused in eSTM40 at Crolles 300, France, and assembled in TSSOP20, VFDFPN8 and UFQFPN32 packages has successfully passed the reliability evaluation performed in agreement to AEC_Q100 Rev.H specification Grade 2.

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1. RELIABILITY STRATEGY

1.1. Reliability strategy

Reliability evaluation is based on the standard STMicroelectronics corporate procedures for quality and reliability, namely RELIABILITY TESTS AND CRITERIA FOR QUALIFICATION, and in compliance with the JESD-47 international standard AEC Q100 rev. H.

The ST33K1M5A is processed in eSTM40 at Crolles 300. This process technology is already qualified.
The ST33K1M5A is assembled in TSSOP20 package at ‘ST SHENZHEN -CHINA’. This package technology is already qualified.
The ST33K1M5A is assembled in VFDFPN8 5x6 WF package at ‘SC UTL1 - UTAC Thai Limited’. This package technology is already qualified.
The ST33K1M5A is assembled in UFQFPN32 5x5 WF package at ‘SC UTL1 - UTAC Thai Limited’. This package technology is already qualified.

In accordance with the applicable ST corporate procedures, the following qualification strategy has been defined:

- Die qualification in TSSOP20 package on three diffusion lots of ST33K1M5A
- Die qualification add-on in VFDFPN8 and UFQFPN32 packages on one diffusion lot
- TSSOP20 package qualification on three assembly lots of ST33K1M5A
- VFDFPN8 package qualification on three assembly lots of ST33K1M5A
- UFQFPN32 package qualification on three assembly lots of ST33K1M5A
- Electromagnetic radiation tests on one diffusion lot of test vehicle from the eSTM40 process technology

Reliability trials performed as part of this reliability evaluation are listed in below Test Plan. For details on test conditions, generic data used and specifications references, refer to test results summary in section 4.

1.2. Test Plan

Table 1 - ***Q100 TEST PLAN***

Test Group	Test name	Description / Comments	Test flag
A Accelerated Environment Stress Tests	**PC**	Preconditioning (JL1 +3 reflows simulation +100 TC*)	Yes
	THB (or HAST)	TemperatureHumidityBias	Yes
	AC (or UHAST) (or THS)	Autoclaveat2atm	Yes
	TC	TemperatureCycling	Yes
	PTC	PowerTemperatureCycling	Not Applicable
	HTSL	HighTemperatureStorageLife	Yes
B Accelerated Lifetime Simulation Tests	**HTOL**	HighTemperatureOperatingLife	Yes
	ELFR	EarlyLifeFailureRate	Yes
	EDR	Electrical Data Retention for NVM	Yes
C Package Assembly Integrity Tests	**WBS**	Wire Bond Shear	Yes
	WBP	WireBondPull	Yes
	SD	Solderability	Yes
	PD	PhysicalDimension	Yes
	SBS	SolderBallShear	Not Applicable
	LI	LeadIntegrity	Not Applicable
E Electrical Verification Tests	**ESD(HBM)**	ElectrostaticDischarge(HumanBodyModel)	Yes
	ESD(CDM)	ElectrostaticDischarge(ChargedDeviceModel)	Yes
	LU	LatchUp	Yes
	ED	Electrical distribution	Yes
	FG	Fault grade	Yes
	CHAR	Characterization	Yes
	EMC	Electromagnetic Compatibility	Yes
	SC	Short Circuit Characterization	Not Applicable
	SER	Soft Error Rate	Yes
	LF	Lead (Pb) Free: (see AEC-Q005)	Yes
D Die Fabrication Reliability Tests	Test list is reported in AEC-Q100 table in section 5	Performed during process qualification	Yes
F Defect Screening Tests	Test list is reported in AEC-Q100 table in section 5	To be implemented starting from first production lot	Yes
G Cavity Package Integrity Tests	Test list is reported in AEC-Q100 table in section 5	N/A: not for plastic packaged devices	Not Applicable

Note: the EMC test have been performed in *[ST33K1M5-A EMC test report v1.1.pdf]*

Table 2 - ***ADDITIONAL TESTS (NO AEC)***

Reason	Test name	Description	Comments
ST internal requirements	**LTOL**	LowTemperatureOperatingLife	Yes
ETSI TS102671	**VVF**	Variable Frequency Vibration	Yes
	MS	Mechanical Shock	Yes
	SA	Salt Atmosphere	Yes
ISO/IEC 10373-1	**UV**	Ultraviolet	Yes
	X-Ray	X-Rays	Yes
	MF	Static magnetic fields	Yes

2. PRODUCT OR TEST VEHICLE CHARACTERISTICS

2.1. Traceability

2.1.1. Wafer Fab Information

Main die – ST33K1M5A

<table>
<tr><th colspan="2">Wafer Fab Information</th></tr>
<tr><th colspan="2">FAB1</th></tr>
<tr><td>Wafer fab name / location</td><td>Crolles 300 / France</td></tr>
<tr><td>Wafer diameter (inches)</td><td>12</td></tr>
<tr><td>Wafer thickness (µm)</td><td>775</td></tr>
<tr><td>Silicon process technology</td><td>CMOSE40</td></tr>
<tr><td>Number of masks</td><td>42</td></tr>
<tr><td>Die finishing front side (passivation) materials</td><td>PSG NITRIDE</td></tr>
<tr><td>Die finishing back side Materials</td><td>RAW SILICON</td></tr>
<tr><td>Die area (Stepping die size)</td><td>4.80 mm² (2614µm, 1837µm)</td></tr>
<tr><td>Die pad size</td><td><table><tr><th>Geometry</th><th>Open(X,Y)</th></tr><tr><td>Rectangular</td><td>120 µm</td></tr><tr><td>Rectangular</td><td>65 µm</td></tr></table></td></tr>
<tr><td>Sawing street width (X,Y) (µm)</td><td>80,80</td></tr>
<tr><td>Metal levels/Materials/Thicknesses</td><td><table><tr><th>Wire bond pad metal</th><th>Composition</th><th>Thickness</th></tr><tr><td>1</td><td>Cu</td><td>0.138 µm</td></tr><tr><td>2</td><td>Cu</td><td>0.163 µm</td></tr><tr><td>3</td><td>Cu</td><td>0.163 µm</td></tr><tr><td>4</td><td>Cu</td><td>0.163 µm</td></tr><tr><td>5</td><td>Cu</td><td>0.163 µm</td></tr><tr><td>6</td><td>Ta/TaN/AlCu</td><td>1.275 µm</td></tr></table></td></tr>
</table>

2.1.2. Assembly Information

Assembly Information	
Package 1: TSSOP 20L BODY 4.40MM LEAD PITCH 0.65MM	
Assembly plant name / location	SHENZHEN B/E / ST SHENZHEN -CHINA
Pitch (mm)	0.65
Die thickness after back-grinding (µm)	280
Die sawing method	Laser grooving + mechanical sawing
Lead frame/Substrate material/supplier/reference	FRAME TSSOP 20L 3x4.20 NiPdAu
Die attach material/type(glue/film)/supplier	GLUE / LOCTITE ABLESTIK
Wire bonding material/diameter/supplier	GOLD / 0.8MILS
Molding compound material/supplier/reference	G700 / SUMITOMO
Package Moisture Sensitivity Level (JEDEC J-STD020D)	1

Assembly Information	
Package 2: VFDFPN 8L 5x6 LEAD PITCH 1.27MM WETTABLE FLANK	
Assembly plant name / location	SC UTL1 - UTAC Thai Limited
Pitch (mm)	1.27
Die thickness after back-grinding (µm)	280
Die sawing method	Laser grooving + mechanical sawing
Lead frame/Substrate material/supplier/reference	L/F DFN8 5x6 Wettable Flanks Matte Sn
Die attach material/type(glue/film)/supplier	GLUE / ABLEBOND
Wire bonding material/diameter/supplier	GOLD / 1MILS
Molding compound material/supplier/reference	G700 / SUMITOMO
Package Moisture Sensitivity Level (JEDEC J-STD020D)	1

Assembly Information	
Package 3: UFQFPN 32L 5x5 LEAD PITCH 0.5MM WETTABLE FLANK	
Assembly plant name / location	SC UTL1 - UTAC Thai Limited
Pitch (mm)	0.5
Die thickness after back-grinding (µm)	127
Die sawing method	Laser grooving + mechanical sawing
Lead frame/Substrate material/supplier/reference	L/F UQFN32 5x5 Wettable Flanks Matte Sn
Die attach material/type(glue/film)/supplier	GLUE / ABLEBOND
Wire bonding material/diameter/supplier	GOLD / 1MILS
Molding compound material/supplier/reference	G700 / SUMITOMO
Package Moisture Sensitivity Level (JEDEC J-STD020D)	1

2.1.3. Reliability testing information

Reliability Testing Information	
Reliability laboratory name / location	Grenoble Rel Lab Rousset MDG Rel Lab

Note: ST is ISO 9001 certified. This induces certification of all internal and subcontractor labs. ST certification document can be downloaded under the following link: http://www.st.com/content/st_com/en/support/quality-and-reliability/certifications.html

3. TEST RESULTS SUMMARY

3.1. Lot information

Lot #	Diffusion Lot / Wafer ID	Assy Lot / Trace Code	Raw Line	Package	Note
Lot 1	VQ052434#10	GK1460MNRN / GK1460MN	52YA*Q4AAGA4	TSSOP 20 BODY 4.4 PITCH 0.65	
Lot 2	VQ104558#06	GK2010RQRP / GK2010RQ	52YA*Q4AAGA4	TSSOP 20 BODY 4.4 PITCH 0.65	
Lot 3	VQ103285#10	GK1490JMRR / GK1490JM	52YA*Q4AAGA4	TSSOP 20 BODY 4.4 PITCH 0.65	
Lot 4	VQ052434#11	2A00983 / FN201064	51BH*Q4AAGA4	VFDFPN 8 5X6 WETTABLE FLANK	
Lot 5	VQ103285#11	2A01777 / FN20207Q	51BH*Q4AAGA4	VFDFPN 8 5X6 WETTABLE FLANK	
Lot 6	VQ104558#11	2A02196 / FN203012	51BH*Q4AAGA4	VFDFPN 8 5X6 WETTABLE FLANK	
Lot 7	VQ104558	2A00983 / FN201064	517M*Q4AAGA4	UFQFPN 32 5X5 WETTABLE FLANK	
Lot 8	VQ101491	2A01777 / FN20207Q	517M*Q4AAGA4	UFQFPN 32 5X5 WETTABLE FLANK	
Lot 9	VQ103285	2A02196 / FN203012	517M*Q4AAGA4	UFQFPN 32 5X5 WETTABLE FLANK	

3.2. Test results summary

For Automotive ICs, test plan results are summarized in AEC-Q100 rev.H template.
Electrical Testing Temperature and Physical Analysis required by AEC-Q100 and any additional STM requirements are also reported in the table below in Test Conditions column.
Test method revision reference is the one active at the date of reliability trial execution.

Tests results summary (Q100 Rev.H)

TEST GROUP A - ACCELERATED ENVIRONMENT STRESS TESTS

Test	#	Reference	Test Conditions	Lots Qty	S.S.	Total	Results/Lot Fail/S.S.	Comments(N/A =Not Applicable)
PC	A1	JESD22- A113 J-STD-020	24h bake@125°C MSL1 3x Reflow simulation Peak Reflow Temp= 260°C ☒ Testing at Room ☒ Testing at Hot ☒ 100x -50/+150deg Temperature Cycles after reflow	3	231	693	TSSOP20 Lot 1 : 0/231 Lot 2 : 0/231 Lot 3 : 0/231	PASS Done before HAST, uHAST and TC
				3	231	693	VFDFPN8 Lot 4 : 0/231 Lot 5 : 0/231 Lot 6 : 0/231	
				3	231	693	UFQFPN32 Lot 7 : 0/231 Lot 8 : 0/231 Lot 9 : 0/231	
HAST	A2	JESD22 A110	P=1.2atm Ta=110°C 85% RH Duration= 264hrs VCC=3.6v ☒ After PC ☒ Testing at Room ☒ Testing at Hot	3	77	231	TSSOP20 Lot 1 : 0/77 Lot 2 : 0/77 Lot 3 : 0/77	PASS
				3	77	231	VFDFPN8 Lot 4 : 0/77 Lot 5 : 0/77 Lot 6 : 0/77	
				3	77	231	UFQFPN32 Lot 7 : 0/77 Lot 8 : 0/77 Lot 9 : 0/77	
uHAST	A3	JESD22 A118	P=1.2atm Ta=110°C 85% RH Duration= 264hrs ☒ After PC ☒ Testing at Room	3	77	231	TSSOP20 Lot 1 : 0/77 Lot 2 : 0/77 Lot 3 : 0/77	PASS
				3	77	231	VFDFPN8 Lot 4 : 0/77 Lot 5 : 0/77 Lot 6 : 0/77	
				3	77	231	UFQFPN32 Lot 7 : 0/77 Lot 8 : 0/77 Lot 9 : 0/77	

TC	A4	JESD22 A104	Ta=-65°C / 150 °C Duration= 500cycles ☒ After PC ☒ Testing at Hot	3	77	231	TSSOP20 Lot 1 : 0/77 Lot 2 : 0/77 Lot 3 : 0/77	PASS
				3	77	231	VFDFPN8 Lot 4 : 0/77 Lot 5 : 0/77 Lot 6 : 0/77	
				3	77	231	UFQFPN32 Lot 7 : 0/77 Lot 8 : 0/77 Lot 9 : 0/77	
HTSL	A6	JESD22 A103	Ta= 150ºC Duration= 1000hrs ☒ Testing at Room ☒ Testing at Hot	3	77	231	TSSOP20 Lot 1 : 0/77 Lot 2 : 0/77 Lot 3 : 0/77	PASS
				3	77	231	VFDFPN8 Lot 4 : 0/77 Lot 5 : 0/77 Lot 6 : 0/77	
				3	77	231	UFQFPN32 Lot 7 : 0/77 Lot 8 : 0/77 Lot 9 : 0/77	

Note: the tests were conducted on product soldered on PCBs.

TEST GROUP B - ACCELERATED LIFETIME SIMULATION TESTS

Test	#	Reference	Test Conditions	Lots Qty	S.S.	Total	Results/Lot Fail/S.S.	Comments(N/A =Not Applicable)
HTOL	B1	JESD22 A108	Ta= 125ºC Duration= 1000hrs VCC=3V6, VDD=1V4 ☒After PC ☒Testing at Room ☒Testing at Cold ☒Testing at Hot	3	77	231	TSSOP20 Lot 1 : 0/77 Lot 2 : 0/77 Lot 3 : 0/77	PASS
				1	77	77	VFDFPN8 Lot 4 : 0/77	
				1	77	77	UFQFPN32 Lot 7 : 0/77	
ELFR	B2	AEC-Q100-008	Ta= 125ºC Duration= 48hrs ☒Testing at Room ☒Testing at Hot	3	800	2400	TSSOP20 Lot 1 : 0/800 Lot 2 : 0/800 Lot 3 : 0/800	PASS
				1	800	800	VFDFPN8 Lot 4 : 0/800	
				1	800	800	UFQFPN32 Lot 7 : 0/800	
EDR	B3	AEC-Q100-005	**NVCE + PCHTDR** *500Kcyc @125°C* *+ 100h bake @150°C* ☒Testing at Room ☒Testing at Hot	3	77	231	Lot 1 : 0/77 Lot 2 : 0/77 Lot 3 : 0/77	PASS
EDR	B3	AEC-Q100-005	**NVCE + HTOL** *500Kcyc @125°C* *+ 1000h HTOL @125°C* ☒Testing at Room ☒Testing at Hot	3	77	231	Lot 1 : 0/77 Lot 2 : 0/77 Lot 3 : 0/77	PASS
EDR	B3	AEC-Q100-005	**NVCE + LTDR** *500Kcyc @25°C* *+ 500h LTDR @25°C* ☒Testing at Room ☒Testing at Hot	3	77	231	Lot 1 : 0/77 Lot 2 : 0/77 Lot 3 : 0/77	PASS

Note: the tests were conducted on product soldered on PCBs.

TEST GROUP C - PACKAGE ASSEMBLY INTEGRITY TESTS

Test	#	Reference	Test Conditions	Lots Qty	S.S.	Total	Results/Lot Fail/S.S.	Comments(N/A =Not Applicable)
WBS	C1	AEC-Q100-001 AEC-Q003	Wire Bond Shear Test: (Cpk > 1.67)	3*10 bonds	5 parts	30 bonds	All measurement within spec limits	PASS on TSSOP20, VFDFPN8 and UFQFPN32
WBP	C2	Mil-STD-883, Method 2011 AEC-Q003	Wire Bond Pull: (Cpk > 1.67); Each bonder used	3*10 bonds	5 parts	30 bonds	All measurement within spec limits	PASS on TSSOP20, VFDFPN8 and UFQFPN32
SD	C3	JESD22 B102 JSTD-002D	Solderability: (>95% coverage) 8hr steam aging prior to testing	1	15	15	All measurement within spec limits	PASS on TSSOP20, VFDFPN8 and UFQFPN32
PD	C4	JESD22 B100, JESD22 B108 AEC-Q003	Physical Dimensions: (Cpk > 1.67)	3	10	30	All measurement within spec limits	PASS on TSSOP20, VFDFPN8 and UFQFPN32

TEST GROUP D - DIE FABRICATION RELIABILITY TESTS

Test	#	Reference	Test Conditions	Lots Qty	S.S.	Total	Results/Lot Fail/S.S.	Comments(N/A =Not Applicable)
EM	D1	JESD61	Data, test method and criteria should be available upon request	-	-	-	PASS	Process qualification data
TDDB	D2	JESD35	Data, test method and criteria should be available upon request	-	-	-	PASS	Process qualification data
HCI	D3	JESD60 & 28	Data, test method and criteria should be available upon request	-	-	-	PASS	Process qualification data
NBTI	D4	JESD90	Data, test method and criteria should be available upon request	-	-	-	PASS	Process qualification data
SM	D5	JESD61, 87, & 202	Data, test method and criteria should be available upon request	-	-	-	PASS	Process qualification data

TEST GROUP E - ELECTRICAL VERIFICATION

Test	#	Reference	Test Conditions	Lots Qty	S.S.	Total	Results/Lot Fail/S.S.	Comments(N/A =Not Applicable)
TEST	E1	User/Supplier Specification	Pre and Post Stress Electrical Test: All parametric and functional tests	All	All	All	PASS	
HBM	E2	AEC-Q100-002	Target HBM= +/-4kV ☒ Testing at Room ☒ Testing at Hot	3	3	9	TSSOP20 Lot 1 : 0/3 Lot 2 : 0/3 Lot 3 : 0/3	PASS
				1	3	3	VFDFPN8 Lot 4 : 0/3	
				1	3	3	UFQFPN32 Lot 7 : 0/3	
CDM	E3	AEC-Q100-011	Target CDM= +/-750V on corner pins; /- 500V all others ☒ Testing at Room ☒ Testing at Hot	3	3	9	TSSOP20 Lot 1 : 0/3 Lot 2 : 0/3 Lot 3 : 0/3	PASS
				1	3	3	VFDFPN8 Lot 4 : 0/3	
				1	3	3	UFQFPN32 Lot 7 : 0/3	
LU	E4	AEC-Q100-004	Current Injection & Overvoltage ☒ Testing at Room ☒ Testing at Hot	3	6	18	TSSOP20 Lot 1 : 0/6 Lot 2 : 0/6 Lot 3 : 0/6	PASS Class II A+
				1	6	6	VFDFPN8 Lot 4 : 0/6	
				1	6	6	UFQFPN32 Lot 7 : 0/6	
ED	E5	AEC-Q100-009 AEC-Q003	Electrical Distributions:(Test @ Rm/Hot/Cold)(where applicable, Cpk >1.67) Corner Lot	-	-	-	PASS	14wafers*13848 = 193872 dies Calculation on Corner Process
FG	E6	AEC-Q100-007	Fault Grading	-	-	-	Fault Grade 90%	PASS Stuckat: 98.86% Atspeed: 87,11% Pseudo stuckat IDDQ: 75.83%
CHAR	E7	AEC-Q003	Characterization:(Test @ Rm/Hot/Cold) Corner Lot	-	-	-	Other (explain)	Report Done All parameters in line with Datasheet
SER	E11	JESD89-1 JESD89-2 JESD89-3	Soft Error Rate	1	3	3		PASS All SER < URD Mission profile : Alpha Soft Error Rate : - STD Cell : < 1000 FIT/Mcell - SRAMs, ROM: < 1250 FIT/Mb - Flash (eSTM) : < 10 FIT/Mb Neutron Soft Error Rate : - STD Cell : < 1000 FIT/Mcell - SRAMs/ROM : < 1250 FIT/Mb - Flash (eSTM) : < 10 FIT/Mb
LF	E12	AEC-Q005	Lead(Pb) Free: (see AEC-Q005)	-	-	-		Covered by Test group A & C

TEST GROUP F - DEFECT SCREENING TESTS

Test	#	Reference	Test Conditions	Lots Qty	S.S.	Total	Results/Lot Fail/S.S.	Comments(N/A =Not Applicable)
PAT	F1	AEC-Q001	Process Average Testing: (see AEC-Q001)	All	All	All	Reject units outside Avg.	Included in the Test and Screening methodology, but postprocessing for wafermap update (defect identification)
SBA	F2	AEC-Q002	Statistical Bin/Yield Analysis: (see AEC-Q002)	All	All	All	Reject units outside criteria	Included in the Test and Screening methodology, but postprocessing for wafermap update (defect identification)

3.2.1. Additional Tests Results Summary

ADDITIONAL TESTS

Test	Reference	Test Conditions	Lots Qty	S.S.	Total	Results/Lot Fail/S.S.	Comments:(N/A =Not Applicable)
LTOL	JESD22-A108	Ta= -40°C Duration= 1000hrs VCC=3V6, VDD=1V4 ☒Testing at Room ☒Testing at Cold ☒Testing at Hot	3	77	231	Lot 1 : 0/77 Lot 2 : 0/77 Lot 3 : 0/77	PASS
UV	ISO/IEC 10373-1	Wavelength 254nm 0.15W/mm² on each side	1	1	1	Lot 1 : 0/1	PASS (1)
X-Ray	ISO/IEC 10373-1	Acceleration 100kV 0,1Gy on each side	1	1	1	Lot 1 : 0/1	PASS (1)
MF	ISO/IEC 10373-1	8000Oe	1	1	1	Lot 1 : 0/1	PASS (1)
VVF	ETSI TS102671 JESD22-B103	Service condition 1 : 20Hz to 2000Hz / Acceleration 20 G ☒Testing at Room	1	15	15	TSSOP20 Lot 1 : 0/15	PASS
			1	15	15	VFDFPN8 Lot 4 : 0/15	
			1	15	15	UFQFPN32 Lot 7 : 0/15	
MS	ETSI TS102671 JESD22-B104	Service condition B : Acceleration 1500 G ☒Testing at Room	1	15	15	TSSOP20 Lot 1 : 0/15	PASS
			1	15	15	VFDFPN8 Lot 4 : 0/15	
			1	15	15	UFQFPN32 Lot 7 : 0/15	
SA	ETSI TS102671 JESD22-A107 / MIL-STD-883 Method 1009.8	Test condition A : 24h ☒Testing at Room	1	15	15	TSSOP20 Lot 1 : 0/15	PASS
			1	15	15	VFDFPN8 Lot 4 : 0/15	
			1	15	15	UFQFPN32 Lot 7 : 0/15	

(1) Test performed on test vehicle ST31P same E40 Front End process technology.

4. APPLICABLE AND REFERENCE DOCUMENTS

Remove from below list not applicable documents.

Reference	Short description
AEC-Q100	Failure Mechanism Based Stress Test Qualification for Integrated Circuits in automotive applications
AEC-Q101	Failure Mechanism Based Stress Test Qualification for Discrete Semiconductors in automotive applications
AEC-Q103	Failure Mechanism Based Stress Test Qualification for Sensors in automotive applications
AEC-Q104	Failure Mechanism Based Stress Test Qualification for Multichip Modules (MCM) in automotive applications
JESD47	Stress-Test-Driven Qualification of Integrated Circuits
SOP2.4.4	Record Management Procedure
SOP2.6.2	Internal Change Management
SOP2.6.7	Finished Good Maturity Management
SOP2.6.9	Package & Process Maturity Management in BE
SOP2.6.11	Program Management for Product Development
SOP2.6.17	Management of Manufacturing Transfers
SOP2.6.19	Front-End Technology Platform Development and Qualification
DMS 0061692	Reliability Tests and Criteria for Product Qualification

5. GLOSSARY

AC	Autoclave	**MR**	Multiple Reflow
ACBV	AC Blocking Voltage	**MS**	Mechanical Shock
ASER	Accelerated Soft Error Rate	**MSeq**	Mechanical sequence
AST	Adhesion Shear Test	**MSL**	Moisture Sensitivity Level
BI	Burn-In	**NVM**	Non Volatile Memory
BT3P	Board 3 points Bending Test	**PC**	Preconditioning
BT4P	Board 4 points Bending Test	**PD**	Physical Dimensions
CA	Constant Acceleration	**PTC**	Power Temperature Cycling
CDM	Electrostatic Discharge – Charged Device Model	**RS**	Repetitive Surge Test
ConA	Construction Analysis	**TSH**	Resistance to Solder Heat
CVS	Constant Voltage Stress	**RTSER**	Real-Time Soft Error Rate
DBT	Dead Bug Test	**SAM**	Scanning Acoustic Microscopy
DPA	Destructive Physical Analysis	**SBP**	Solder Ball Pull
DROP	Package drop	**SBS**	Solder Ball Shear
DS	Die Shear	**SC**	Short Circuit Characterization
DToB	Drop Test on Board	**SCCSS**	Smartcard – Constant Supply Stress
EDR	NVM Program/Erase Endurance & Data Retention Stress Test	**SCMCMS**	Smartcard – MasterCard Mechanical Stress
ELFR	Early Life Failure Rate	**SCMF**	Smartcard – Magnetic Field Stress
EMC	Electromagnetic Compatibility	**SCPOOS**	Smartcard – Power Off/On Stress
EOS	Electrical Overstress characterization	**SCRFC**	Smartcard – RF On/Off Cyclic Stress
ESeq	Environmental sequence	**SCRFS**	Smartcard – RF On Static Stress
EV	External Visual	**ScrT**	Screw Test
FF	Free Fall	**SCSA**	Smartcard – Salt Atmosphere
GFL	Gross/Fine Leak	**SCUV**	Smartcard – UV Test
GL	Electro-thermally Induced Gate Leakage	**SCXRAY**	Smartcard – XRAY Test
GStress	Gate Stress	**SD**	Solderability
GUN	Electrostatic Discharge - System Level Test	**SSOP**	Steady State Operational
H3TRB	High Humidity High Temperature Reverse Bias	**SToB**	Shock Test on Board
HAST	Biased HAST (Highly Accelerated Stress Test)	**TC**	Temperature Cycling
HBM	Electrostatic Discharge - Human Body Model	**TCDT**	Temperature Cycling Delamination Test
HER	Hermeticity	**TCHT**	Temperature Cycling Hot Test
HMM	Electrostatic Discharge - Human Metal Model	**TCoB**	Temperature Cycling on Board
HTFB	High Temperature Forward Bias	**THB**	Temperature Humidity Bias
HTGB	High Temperature Gate Bias	**THS**	Temperature Humidity Storage
HTHHB	High Temperature High Humidity Bias	**TLP**	Electrostatic Discharge – Transmission Line Pulse
HTOL	High Temperature Operating Life	**TS**	Thermal Shocks
HTRB	High Temperature Reverse Bias.	**TStr**	Terminal Strength
HTSL	High Temperature Storage Life	**Tumb**	Tumbler Test
IOL	Intermittent Operating Life	**UHAST**	Unbiased HAST (Highly Accelerated Stress Test)
IWV	Internal Water Vapor	**VToB**	Vibration Test on Board
LF	Lead Free	**VFV**	Variable Frequency Vibration
LI	Lead Integrity	**WAT**	Tin (Sn) Whisker Acceptance Testing
LT	Lid Torque	**WBI**	Wire Bond Integrity
LTOL	Low Temperature Operating Life	**WBP**	Wire Bond Pull
LTSL	Low Temperature Storage Life	**WBS**	Wire Bond Shear
LU	Latch-Up	**WBSt**	Wire Bond Strength
MM	Electrostatic Discharge - Machine model	**XRAY**	X ray inspection

6. REVISION HISTORY

Release	Date	Description
1.0	07/02/2023	Initial release
1.1	Feb 23, 2023	UFQFPN32 Typo correction
1.2	April 23, 2024	Chapter 3: Added text " Note: the tests were conducted on product soldered on PCBs" for **TEST GROUP A** and **B sections**
1.3	July 1rst, 2024	Chapter3: Updated LU quantity to 6. Added conditions for VVF, MS and SA
1.4	Sept 1rst, 2024	Removed "ST Restricted"
1.5	Sept 27th, 2024	Additional info in "Test Conditions" column for Group A and B tables
1.6	Feb 4th, 2025	SER result and EMC report added.
1.7	Feb 13th, 2025	HBM level updated

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